

X-ray imaging requires the use of harmful ionizing radiation to image structures inside the body. An x-ray image intensifier, shown in Figure 1, allows high-quality imaging with very low radiation exposure to the patient.

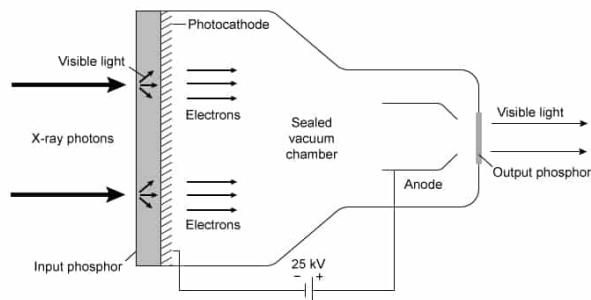


Figure 1 Schematic of an x-ray image intensifier

The x-ray image intensifier amplifies the imaging signal by approximately 300,000 times. The intensifier converts the image data from x-ray photons to visible light, then to electrons, and back to visible light. At each conversion stage, the signal is amplified up to 1,000 times.

The x-ray image intensifier consists of a vacuum chamber that contains an input phosphor, a photocathode, focusing electrodes (not shown in Figure 1), an anode, and an output phosphor. The anode voltage is 25 kV higher than the photocathode, generating a strong electric field in the vacuum. The input phosphor converts x-ray photons into visible light, which then ejects electrons from the photocathode. Even with an anode voltage of 25 kV, no current will flow through the vacuum unless electrons are ejected from the photocathode. Ejected electrons are accelerated toward the anode and focused onto the output phosphor. At the output phosphor, the electrons are converted back into visible light, allowing image data to be recorded with a camera or video system. (Note: Use $c = 3 \times 10^8$ m/s in a vacuum, if needed.)

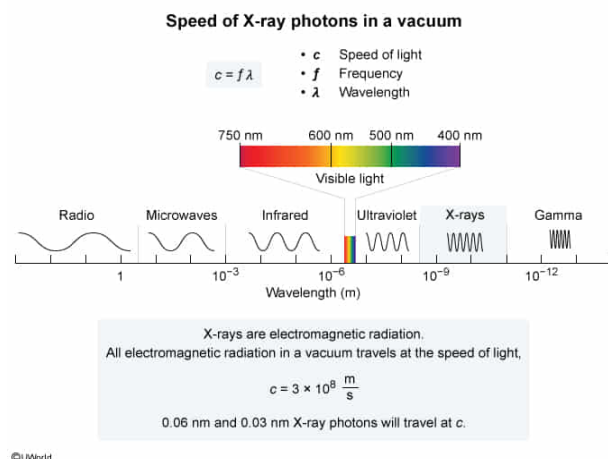
Two x-ray photons with wavelengths of 0.06 nm and 0.03 nm are traveling in a vacuum. How will the speed of the 0.06 nm x-ray photon compare with the speed of the 0.03 nm x-ray photon?

- ☐ A. It will be 1/4 as fast [2%]
- ☐ B. It will be 1/2 as fast [26%]
- ✓ ☒ C. It will be equal [43%]
- ☐ D. It will be 2 times faster [27%]

Correct

42% answered correctly

Explanation:



X-rays are a form of **electromagnetic radiation** with **wavelengths** λ ranging from approximately 0.01 nm to 1 nm. In a vacuum, all electromagnetic radiation has the same **speed** c , equal to the product of the frequency f and λ :

$$c = f \lambda$$

As the value of λ increases for different types of x-rays, the value of f decreases by an identical proportion such that c remains constant at 3×10^8 m/s.

In this question, the speed of a 0.06 nm x-ray photon is compared to the speed of a 0.03 nm x-ray photon in a vacuum. Because the speeds of all x-rays in a vacuum equal c , the 0.06 nm and 0.03 nm x-rays have the same speed.

(Choices A, B, and D) All x-rays are electromagnetic radiation and have the same speed in a vacuum, c (ie, 3×10^8 m/s). X-rays can have different values for λ , but their speed is always c .

Educational objective:

X-rays are electromagnetic radiation and have a speed in a vacuum equal to the speed of light, independent of the wavelength of the x-rays.

Time spent: 0 sec

Subject:
Physics

QID: 403352

System:
Light and Sound

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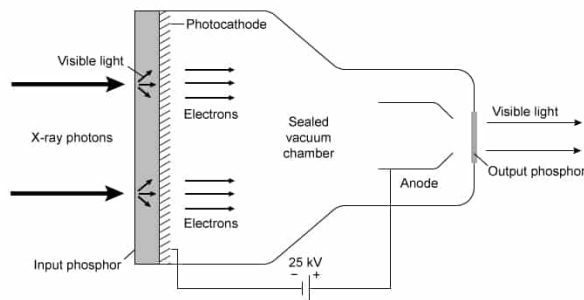


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Visible light travels through the output phosphor at 1.6×10^8 m/s. What is the refractive index of the material?

- ✓ ☒ A. 1.9 [67%]
☐ B. 1.1 [6%]
☐ C. 0.88 [7%]
☐ D. 0.53 [18%]

Correct

66% answered correctly

Explanation:

Refractive index of output phosphor

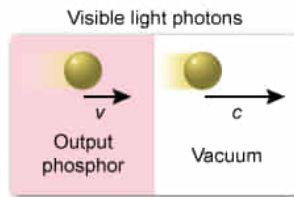
$$n = \frac{c}{v}$$

$$c = 3 \times 10^8 \frac{\text{m}}{\text{s}}$$

- n Refractive index
- c Speed in vacuum
- v Speed in phosphor

$$v = 1.6 \times 10^8 \frac{\text{m}}{\text{s}}$$

$$n = \frac{c}{v} = \frac{3 \times 10^8 \frac{\text{m}}{\text{s}}}{1.6 \times 10^8 \frac{\text{m}}{\text{s}}} \approx 1.9$$



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Visible light is **electromagnetic radiation** with wavelengths ranging from 400 nm to 750 nm. All electromagnetic radiation has the same **speed** of c in a vacuum (ie, 3×10^8 m/s). However, when electromagnetic radiation enters a medium—such as air, water, or a transparent solid—its speed changes. The **index of refraction** n of the material equals the ratio of c to the speed of light v in the medium:

$$n = \frac{c}{v}$$

In this question, the value of v in the output phosphor is given as 1.6×10^8 m/s. Therefore, the value of n for this material is:

$$n = \frac{3 \times 10^8 \frac{\text{m}}{\text{s}}}{1.6 \times 10^8 \frac{\text{m}}{\text{s}}} \approx 1.9$$

(Choice B) The value of 1.1 is obtained by incorrectly calculating the ratio of v and $c - v$.

(Choice C) The value of 0.88 is obtained by incorrectly calculating the ratio of $c - v$ and v .

(Choice D) The value of 0.53 is obtained by incorrectly calculating the ratio of v and c , which is the inverse of n .

Educational objective:

The refractive index of a material equals the ratio of the speed of light in a vacuum to the speed of light in the material.

Time spent: 0 sec

Subject:
Physics

QID: 403353

System:
Light and Sound

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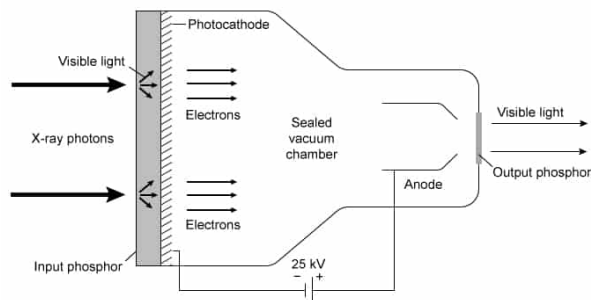
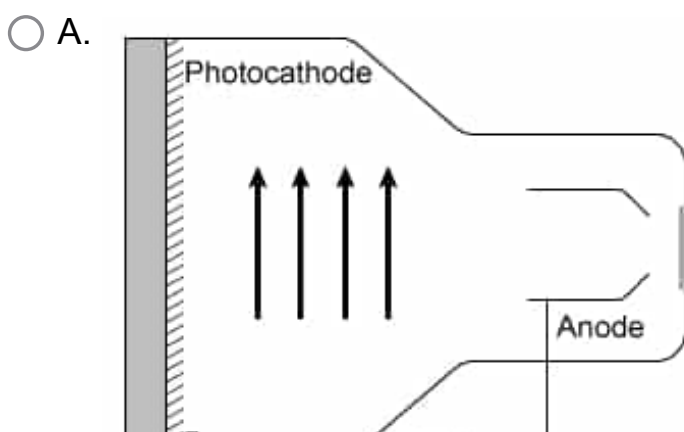


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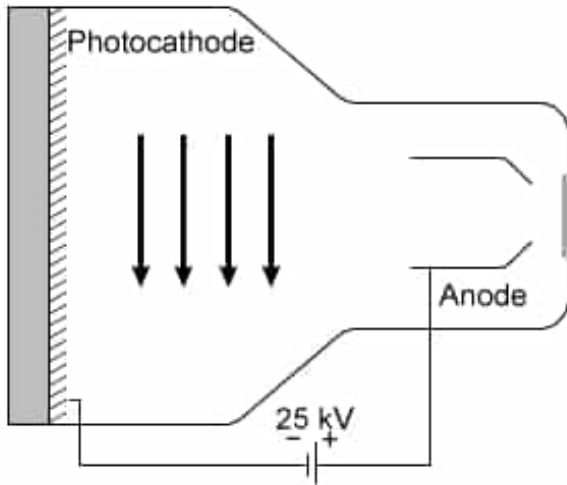
Which of the following figures best illustrates the direction of the electric field lines between the photocathode and anode of an x-ray image intensifier?





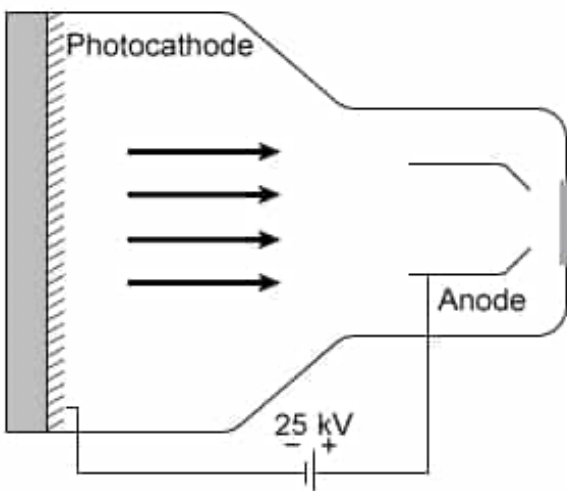
[14%]

☐ B.



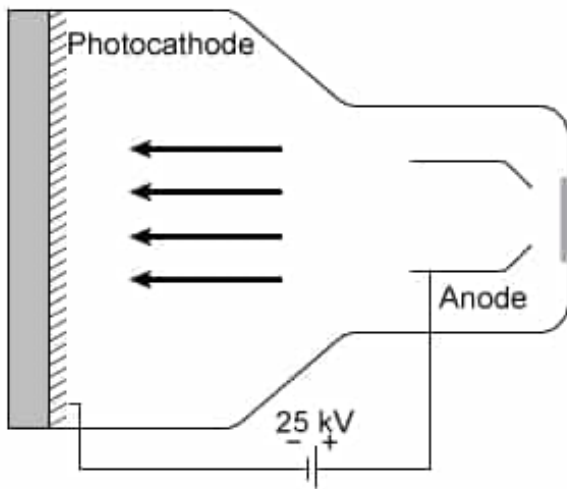
[8%]

☐ C.



[32%]

☒ D.



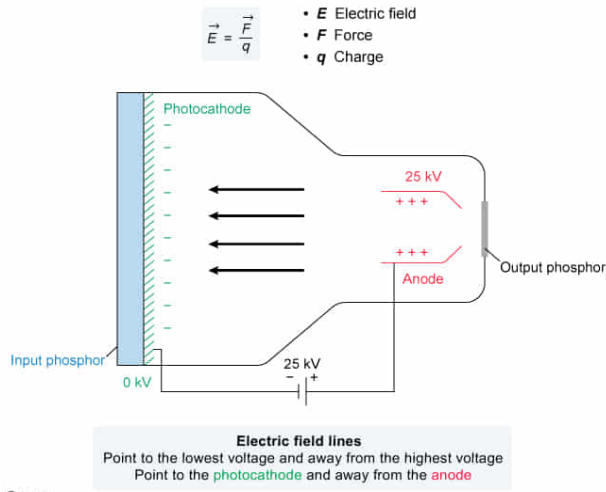
[44%]

Correct

45% answered correctly

Explanation:

Electric field lines of an X-ray image intensifier



The **electric field** E equals the ratio of the electrostatic force F experienced by a charged particle within the field to the charge q of the particle:

$$\vec{E} = \frac{\vec{F}}{q}$$

E and F are vectors, and therefore they each have both a direction and a magnitude. The **sign of q** determines if the directions of F and E are identical or opposite each other.

Electric field lines graphically indicate the direction and relative strength of E . Electric field lines begin at positive charges and end at negative charges; hence, they point away from regions of higher voltages and toward regions of lower voltages. E induces positive charges to move in the same direction as the field lines and induces negative charges to move in the opposite direction.

In this question, the electric field is generated by the **photocathode** and **anode** voltages. The photocathode voltage is 0 V and the anode voltage is 25 kV. Therefore, the electric field lines should point toward the photocathode and away from the anode, corresponding to the figure shown in Choice D.

(Choices A and B) In these figures, the electric field lines are not in the same direction as the electrostatic force on electrons. Furthermore, the lines do not point away from higher voltages and toward lower voltages.

(Choice C) The electric field lines here point toward the anode, but they should point toward the photocathode.

Educational objective:

Electric field lines denote the direction and relative strength of the field. The field lines point toward the lowest voltage in the field and away from the highest voltage.

Time spent:0 sec**Subject:**

Physics

QID:403354**System:**

Light and Sound

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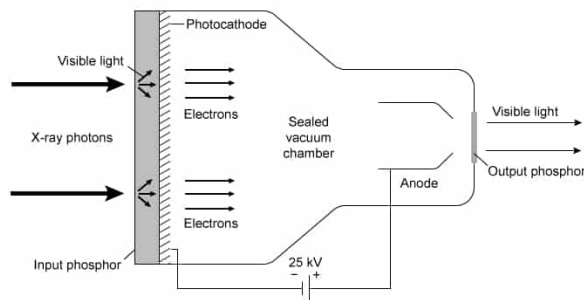


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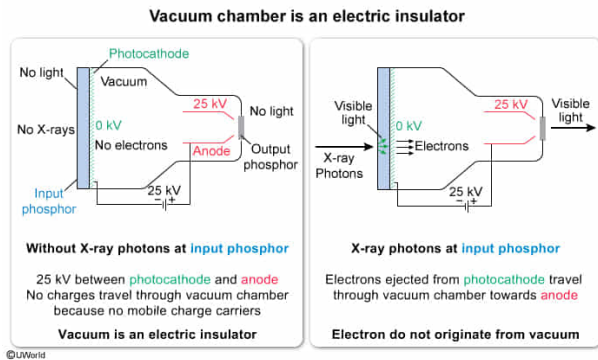
Is the vacuum chamber in an x-ray image intensifier an electrical insulator?

- ✔ ☒ A. Yes, because only electrons ejected from the photocathode can travel through the vacuum chamber [70%]
- ☐ B. Yes, because no electrons can travel through the vacuum chamber [6%]
- ☐ C. No, because electrons can travel through the vacuum chamber even without x-rays at the input phosphor [18%]
- ☐ D. No, because electrons in the vacuum chamber are not exposed to a voltage difference [4%]

Correct

70% answered correctly

Explanation:



Electrical **conductors** allow the flow of electric charges when they are exposed to a voltage because they contain mobile charge carriers, such as negatively charged electrons or positively charged carriers (often called "holes"). Electrical **insulators** do not contain mobile charge carriers, and thus do not allow electric charges to flow when a voltage is applied.

In this question, the vacuum chamber of an x-ray image intensifier is exposed to 25 kV. The **vacuum** chamber is formed by removing all gases present, leaving almost no matter within the chamber (and no mobile charge carriers). As a result, no charges can travel through the vacuum chamber with only the applied voltage. However, electrons ejected from the photocathode can enter the vacuum chamber. These electrons experience a voltage of 25 kV inside the vacuum and accelerate toward the anode.

Therefore, the vacuum chamber is an electrical insulator because it does not contain any charge carriers, and only electrons ejected from the photocathode can travel through the vacuum chamber.

(Choice B) The vacuum chamber is an electrical insulator, but electrons ejected from the photocathode can travel through the vacuum chamber.

(Choice C) The vacuum chamber is an electrical insulator because electrons cannot travel through the vacuum chamber without x-rays at the input phosphor. Electrons are ejected from the photocathode only when x-rays are converted into visible light by the input phosphor.

(Choice D) The vacuum chamber is an electrical insulator. Furthermore, electrons in the vacuum chamber are exposed to a voltage difference of 25 kV from the photocathode and the anode.

Educational objective:

Electrical insulators do not contain charge carriers, and electric charges will not flow with only an applied voltage. A vacuum is an electrical insulator because it does not contain charge carriers.

Time spent:0 sec

Subject:

Physics

QID:403355

System:

Light and Sound

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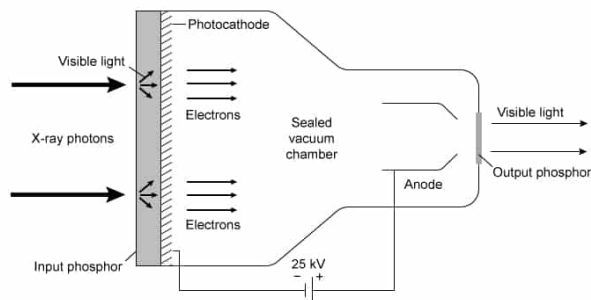


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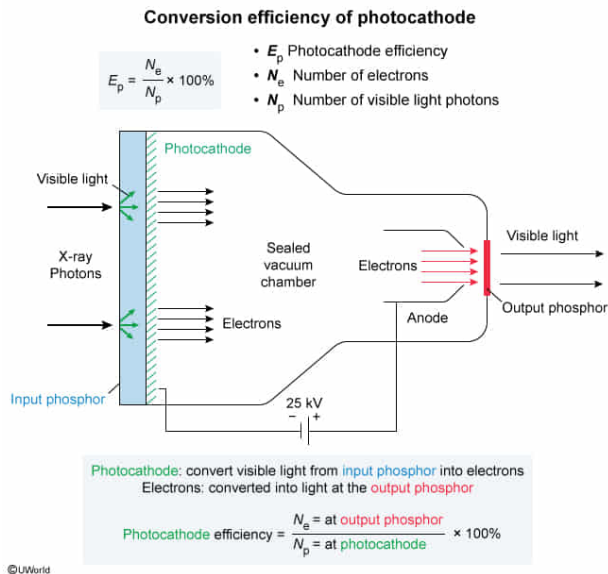
Researchers want to measure the efficiency of the photocathode for converting visible light into electrons. Which of the following measurements should they make?

- ☐ A. Number of x-rays incident on the input phosphor and number of visible light photons generated at the output phosphor [12%]
- ☐ B. Number of visible light photons incident on the output phosphor and number of electrons incident on the photocathode [12%]
- ☒ C. Number of visible light photons incident on the photocathode and number of electrons incident on the output phosphor [50%]
- ☐ D. Number of visible light photons incident on the photocathode and number of electrons generated by the input phosphor [24%]

Correct

50% answered correctly

Explanation:



The overall function of an x-ray image intensifier is to convert and amplify x-rays into visible light. This result is achieved through several intermediate steps: X-rays are first converted into visible light **photons** by the input phosphor, which are converted into **electrons** by the **photocathode**, and finally the electrons are converted back into visible light by the output phosphor. To investigate the conversion properties for a single step, its input and output must be isolated from the other steps.

In this question, the input to the photocathode is visible light photons generated in the input phosphor. The output of the photocathode is electrons that accelerate toward the output phosphor. The photocathode efficiency E_p in percent equals the ratio of the number of electrons N_e at the output phosphor to the number of visible light photons N_p at the photocathode multiplied by 100:

$$E_p = \frac{N_e}{N_p} \times 100\%$$

Therefore, E_p can be determined by measuring the number of visible light photons incident on the photocathode and the number of electrons incident on the output phosphor.

(Choice A) These measurements depend on the conversion properties of the input phosphor, photocathode, and output phosphor, not just the photocathode.

(Choice B) Electrons, not visible light photons, are incident on the output phosphor. Furthermore, visible light photons (not electrons) are incident on the photocathode.

(Choice D) The input phosphor generates visible light photons, not electrons.

Educational objective:

An x-ray image intensifier performs a series of conversion and amplification steps.

Determining the efficiency of one particular step requires measurements of the input and output for that individual step.

Time spent:0 sec

Subject:
Physics

QID:403356

System:
Light and Sound

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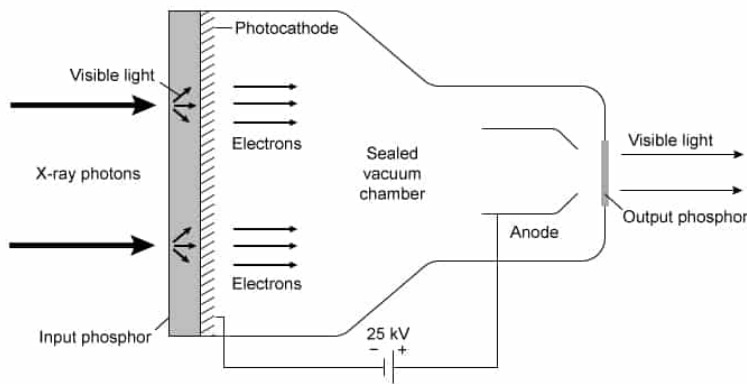


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An image intensifier has a photocathode-to-anode distance of d and an anode voltage of V . An electron with a mass of m and charge of q is ejected from the photocathode with negligible initial kinetic energy. What is the speed of the electron immediately before it reaches the anode?

- ☐ A. $\sqrt{\frac{2qV}{dm}}$ [40%]
- ✓ ☒ B. $\sqrt{\frac{2qV}{m}}$ [43%]
- ☐ C. $\frac{2qV}{dm}$ [11%]
- ☐ D. $\frac{2qV}{m}$ [3%]

Correct

43% answered correctly

Explanation:

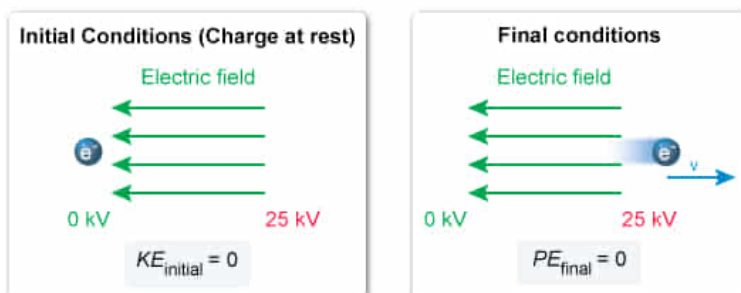
Speed of an electron in an electric field:
Conversion of potential energy into kinetic energy

$$PE_{\text{initial}} = qV$$

$$KE_{\text{final}} = \frac{1}{2} mv^2$$

$$v = \sqrt{\frac{2KE_{\text{final}}}{m}}$$

- PE_{initial} Initial potential energy
- q Charge of electron
- V Voltage
- KE_{final} Final kinetic energy
- m Mass of electron
- v Speed



$$PE_{\text{initial}} = KE_{\text{final}} = qV, v = \sqrt{\frac{2qV}{m}}$$

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For electric charges, the law of **conservation of energy** states that

the total energy E , which is the sum of the **electric potential energy** PE and **kinetic energy** KE of the charge, must remain constant:

$$E = PE + KE = \text{constant}$$

For example, an electron at rest in an electric field has a PE equal to the product of the electron charge q and the voltage V of the field:

$$PE = qV$$

The electric field imparts an **electrostatic force** on the electron, causing it to accelerate. The KE of the moving electron is half the product of its mass m and its speed v squared:

$$KE = \frac{1}{2}mv^2$$

Consequently, v can be expressed in terms of KE and m by rearranging the above equation:

$$v = \sqrt{\frac{2KE}{m}}$$

The E of the electron remains constant, but it is transformed from PE into KE as it accelerates within the electric field, hence:

$$E_{\text{initial}} = E_{\text{final}}$$

$$PE_{\text{initial}} + KE_{\text{initial}} = PE_{\text{final}} + KE_{\text{final}}$$

In this question, the electron initially has negligible KE , so $KE_{\text{initial}} \approx 0$. Consequently, E_{initial} is equal to PE_{initial} :

$$E_{\text{initial}} = PE_{\text{initial}} = qV$$

Immediately before the electron reaches the anode, PE_{final} is zero.

Hence, E_{final} equals KE_{final} :

$$E_{\text{final}} = KE_{\text{final}}$$

Because the PE decreased from qV to zero, KE_{final} must equal qV :

$$KE_{\text{final}} = qV$$

Substituting this result for KE_{final} into the above equation for v yields:

$$v = \sqrt{\frac{2KE_{\text{final}}}{m}}$$

$$v = \sqrt{\frac{2qV}{m}}$$

(Choice A) This equation results from incorrectly calculating PE_{initial} as $\frac{qV}{d}$.

(Choices C and D) These equations are both missing the square root operator required to calculate v from the KE equation.

Educational objective:

The energy of an electron in an electric field is converted from potential energy into kinetic energy. The final speed of the electron depends on its initial potential energy and mass.

Time spent: 0 sec

Subject:

Physics

QID: 403357

System:

Light and Sound